Complex Numbers: Quadratic Equations

Questions 1–12

Question 1. Bruce is attempting to find the solutions to an equation. His attempt is shown below.

$$\begin{aligned} z^2 - 2z + 5 &= 0 \\ (z-2)^2 - 4 &= -5 \\ z - 2 &= \sqrt{-9} \\ z &= 2 \pm 9i \end{aligned}$$

a) Identify all his mistakes.
b) Write a correct solution.

Question 2. Without using a calculator, find the solutions to each of the following equations. You must show your working:

a) $z^2 + 1 = 0$
b) $z^2 + 10 = 0$
c) $z^2 - 2z + 17 = 0$
d) $4z^2 - 4z + 17 = 0$
e) $2z^2 - 2z + 1 = 0$
f) $z^2 - z + 3 = 0$

Question 3. Find the solutions to each of the following, giving your answer in the form $a + bi$:

a) $(z + 5)^2 = -7$

b) $(z - 1)^2 = -12$

c) $2(z + 4)^2 + 8 = 6$

d) $8(z + 5)^2 - 3 = -9$

Question 4. Without using a calculator, find all the solutions to

$$z^4 + 2z^2 - 3 = 0$$

You must show your working.

Question 5. Find the quadratic equation in the form

$$z^2 + bz + c = 0$$

which has roots:

a) $1 + i$ and $1 - i$
b) $3 + i$ and $3 - i$
c) $1 + 2i$ and $1 - 2i$
d) $-4 + 3i$ and $-4 - 3i$
e) i and $-i$
f) $5i$ and $-5i$
g) $-1 + \sqrt{2}i$ and $-1 - \sqrt{2}i$

Question 6. One root the quadratic equation

$$z^2 + pz + q = 0$$

is $3 - 8i$, where p and q are real values.

a) Write down the other root of the equation.
b) Find the values of p and q

Question 7. One root the quadratic equation

$$z^2 + pz + q = 0$$

is $-\frac{1}{2} - 4i$, where p and q are real values.

a) Write down the other root of the equation.

b) Find the values of p and q

Question 8. One root the quadratic equation

$$z^2 + pz + q = 0$$

is $a + bi$, where p, q, a and b are real values.

a) Write down the other root of the equation.

b) Find the values of p and q in terms of a and b

Question 9. Prerequisite knowledge for Q9: Complex Loci

The function f is defined as

$$\mathrm{f}(z) = z^2 - 4z + 7$$

a) Find the roots of $\mathrm{f}(z) = 0$

The two of the roots of $\mathrm{f}(z) = 0$, z_1 and z_2 lie on the locus $C = \{z : |z| = p\}$

b) Find the value of p

c) Find the area of the region in the Argand diagram enclosed by the locus C

Another point w lies on the locus C such that $\mathrm{Re}(w) = -\mathrm{Re}(z_1)$

d) State the possible values of w

The function g is defined as

$$\mathrm{g}(w) = w^2 + aw + b$$

where a and b are real constants.

e) Using your answers to part d) as the roots of $\mathrm{g}(w) = 0$, determine the values of a and b

Question 10. The equation

$$z^2 + kz + 4 = 0$$

where k is a real number, has complex roots. Find the range of possible values of k

Question 11. It is given that a quadratic polynomial has non-zero real coefficients, and the linear term is zero. Show that it has purely imaginary roots.

Question 12. It is given that a quadratic polynomial has non-zero real coefficients, and the constant term is zero. Show that it can only have real roots.